CLOCKS

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↳ The main purpose of knowing clocks concept is that in real time scenarios while we are observing clocks (or) watches, how many times min & hands are coinside, with how much angle the hands are seperated @ a particular time .... etc".

↳ To find all the above first we need to know about Speeds of all moving objects in clock.

↳ The dial of a Clock (or) Circumference of dial of a clock is divided into 60 equal divisions (or) minute spaces (or) minute divisions.

↳ Generally clock has 3 hands, But when this Concept was introduced there were only 2 hands. So, we are considering 2 hands only. minutes & hours hands.

↳ To find out no. of coincidences (or) time taking to overtake the slower hand faster hand we must know the speeds of hands.

↳ Minute hand Speed:- In complete revolution by minutes hand, it can cover 360° of dist in a time of 60 minutes.

60min/1hr ⟶ 360°

1min ⟶ 360/60 = 6°

Speed of minutes hand is 6° per minute

$S_m = 6^\circ$ per minute

↳ Hours hand Speed:-

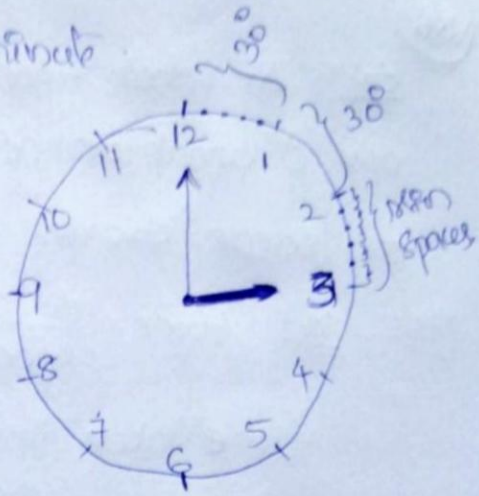
Hours hand can cover 360° in 12 hr's of time.

12 hrs $\longrightarrow 360^\circ$

1 hr = 60 min $\longrightarrow \frac{360^\circ}{12} = 30^\circ$

1 min $\longrightarrow \frac{30^\circ}{60} = \frac{1}{2}^\circ$

Speed of hours hand $= 0.5^\circ$ per minute.



5 min spaces $= 30^\circ$

1 min space $= 60^\circ$

→ To find cross each other / overtakes the slower by faster we must know relative speed.

↳ here two objects are moving in same direction with speeds 0.5° per min & 6° per min. The relative speed is

Relative Speed $= S_1 - S_2 = 6^\circ - 0.5^\circ = 5.5^\circ$ per minute

when slower object is kept as constant.

↳ Relative speed is 5.5° per minute when hr's hand as constant.

* The angular difference between the hands:-(θ):-

Angular difference $(\theta) = |30H - \frac{11}{2}M|$.

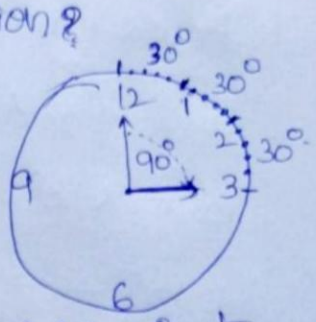
H = No. of hours moved.

M = No. of minutes moved.

Coincidence nothing but $\theta = 0$ [touching each other]

Ex-1:- At what time between 3'o clock & 4'o clock hands are of apart (or) coincide each other (or) touch (or) meet (or) on same straight line in same direction?

Ans:- M-1:- time = $\frac{\text{distance}}{\text{Speed}} = \frac{\text{dist}}{\text{Relative speed}}$



time taken to touch (or) overtake.

$\Rightarrow t_{o/t/m} = \frac{\text{dist travelled in degrees}}{\text{Speed in degrees per minute}}$

$= \frac{90^\circ}{5.5^\circ} = \frac{90\times 2}{5.5\times 2} = \frac{180}{11} = 16\frac{4}{11}$ mins.

Means after $16\frac{4}{11}$ mins to 3'o clock hands are coincide $= 3 + 16\frac{4}{11} = 3$ hr's $16\frac{4}{11}$ minutes am/pm.

M-2:- $\theta = 0 \Rightarrow |30H - \frac{11}{2}M| = 0$

$30H = \frac{11}{2}M$ [H = 3 hr's] [M = ?]

$M = \frac{30\times 3\times 2}{11} = \frac{180}{11} = 16\frac{4}{11}$ minutes.

M-3:- Formula method but very easy:-

time to coincide = $\frac{\text{travelling dist in min spaces}}{\text{travelling dist in min spaces}} \over 11$

$= 15\frac{15}{11} = (15+1)\frac{4}{11} = 16\frac{4}{11}$ minutes

After $16\frac{4}{11}$ minutes hands are coincide each other.

Required time is 3 am/pm + $16\frac{4}{11}$ minutes

$= 3:16\frac{4}{11}$ am/pm.

(149)

Total coincidences in 12 hr's:-

Between 1&2 → 1 hr → 5 min spaces → $5\frac{5}{11}$ → $1:5\frac{5}{11}$

2&3 → 2 hr's → 10 min spaces → $10\frac{10}{11}$ → $2:10\frac{10}{11}$

3&4 → 3 hr's → 15 m.s → $15\frac{15}{11}$ → $3:15\frac{15}{11} = 3:16\frac{4}{11}$.

4&5 → 4 hr's → 20 m.s → $20\frac{20}{11}$ → $4:20\frac{20}{11} = 4:21\frac{9}{11}$.

5&6 → 5 hr's → 25 m.s → $25\frac{25}{11}$ → $5:25\frac{25}{11} = 5:27\frac{3}{11}$.

6&7 → 6 hr's → 30 m.s → $30\frac{30}{11}$ → $6:30\frac{30}{11} = 6:32\frac{8}{11}$.

7&8 → 7 hr's → 35 m.s → $35\frac{35}{11}$ → $7:35\frac{35}{11} = 7:38\frac{2}{11}$.

8&9 → 8 hr's → 40 m.s → $40\frac{40}{11}$ → $8:40\frac{40}{11} = 8:43\frac{7}{11}$.

9&10 → 9 hr's → 45 m.s → $45\frac{45}{11}$ → $9:45\frac{45}{11} = 9:49\frac{1}{11}$.

10&11 → 10 hr's → 50 m.s → $50\frac{50}{11}$ → $10:50\frac{50}{11} = 10:54\frac{6}{11}$.

11&12 → 11 hr's → 55 m.s → $55\frac{55}{11}$ → $11:55\frac{55}{11} = 11:60 = 12$

not b/w 11&12 ← No coincidence. ← Exception.

12&1 → 12 hr's → 60 mins → $60\frac{60}{11}$ → $12:60\frac{60}{11} = 12:65\frac{5}{11}$

= 12 hr's + 60 mins + $5\frac{5}{11}$ = $1:5\frac{5}{11}$ [b/w 1&2]

[which is beyond 1 not b/w 12&1]

→ Exception. → No coincidence

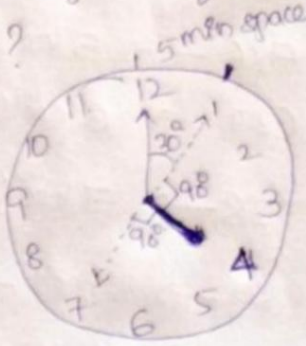
* b/w 11&12 we will be getting at 12'o clock. which is super case [which is not b/w any 2 consecutive hr's & at a particular hr].

Total:- b/w 11 consecutive hr's 1&2, 2&3 ..... 10&11. total 10 hr's gap total 10 times 1 time each and at exactly 12'o clock super case once. Total '11' times in 12 hr's.

(150)

Note:- B/w 1202, 2203 .... 1221, hands are coincidence only once except b/w 11212 and 12221 [Exceptions of 0 coincidence]. Total in a day hands are coincidence [11+11=22] 22 times including 12'o clock [super case] 2 times in a day.

Ex:-2:- After how much time hands are right angles to each other b/w 4'o clock & 5'o clock?



Ans- M-1:- $t_1 = \frac{d_1}{R.S} = \frac{30^\circ}{5.5} = \frac{30\times 2}{5.5\times 2}$

$= \frac{60}{11} = 5\frac{5}{11}$ mins.

$t_2 = \frac{d_2}{R.S} = \frac{(120^\circ+90^\circ)}{5.5} = \frac{210}{5.5}$

$= \frac{210\times 2}{5.5\times 2} = \frac{420}{11} = 38\frac{2}{11}$ min.

M-2:- $\theta = 90^\circ = |30H - \frac{11}{2}M|$, H = 4 hr's

$\Rightarrow \therefore 30H - \frac{11}{2}M = 90$ & $\frac{11}{2}M - 30H = 90$

$\frac{11}{2}M = 30\times 4 - 90$ | $\frac{11}{2}M = 30H + 90 = 30\times 4 + 90$

$\Rightarrow M = \frac{30\times 2}{11} = 5\frac{5}{11}$ | $M = \frac{210\times 2}{11} = 38\frac{2}{11}$ mins

M-3:-

time = travelling dist in min spaces / travelling dist in min spaces in 11.

$t_1 = 5[30^\circ]\frac{5}{11}$, $t_2 = 35\frac{35}{11}[210^\circ] = 38\frac{2}{11}$.

After travelling of $5\frac{5}{11}$ min & $38\frac{2}{11}$ mins b/w 4 & 5 Hands are maintaining 90° angle twice in 1hr gap.